

entanglement generation rate (pairs/s), fidelity, secret-key rate (bits/s), and outage probability under realistic field noise. In contrast to the conventional internet, which is fundamentally vulnerable to eavesdropping and hacking, the quantum internet promises intrinsic security rooted in the laws of physics.

By enabling the direct transmission of quantum information—qubits—across vast distances, this network could facilitate new forms of communication, computation, and sensing. The transformative potential of the quantum internet extends to secure government communications, advanced scientific collaboration, and even the interconnection of quantum computers, thus redefining the boundaries of technological progress.

Entanglement distribution across networks

Entanglement produces correlations between measurement outcomes that are stronger than any classical model; however, these correlations cannot be used for superluminal signalling and become operationally useful only when combined with classical communication and agreed measurement settings. This property forms the backbone of quantum networking, enabling the distribution of entangled pairs across a network to establish secure links. Entanglement distribution is achieved through various methods, including photon transmission over optical fibres and satellite links. The ability to share entangled states between remote nodes allows for the creation of quantum channels, which are essential for quantum key distribution and other secure communication protocols. As entanglement is inherently fragile, maintaining its integrity over long distances remains a key challenge in quantum networking.

In fibre links, attenuation and noise impose hard scaling limits on direct transmission; in free-space and satellite links, pointing, atmospheric turbulence, and background light dominate error budgets. Engineering-grade entanglement distribution therefore relies on tight polarisation/phase stabilisation, spectral filtering, and precise time-tagging to preserve interference visibility at the detectors.

Quantum repeaters and network security

One of the primary obstacles in constructing large-scale quantum networks is the attenuation and decoherence of quantum signals over distance. Quantum repeaters are specialised devices designed to overcome this challenge by extending the range of entanglement and facilitating error correction. Repeaters are typically categorised as: (i) purification-based repeaters, which trade additional entanglement attempts for higher fidelity via distillation, and (ii) error-corrected repeaters, which encode logical qubits to tolerate loss and operational errors at the cost of substantially higher qubit and control overhead. The design choice depends on target distance, required



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